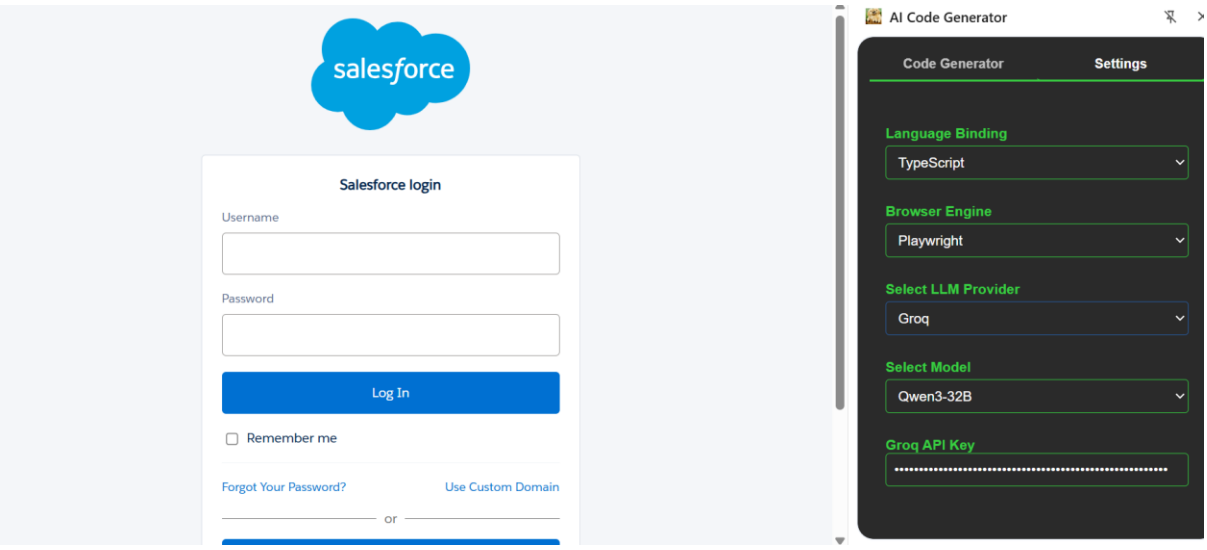

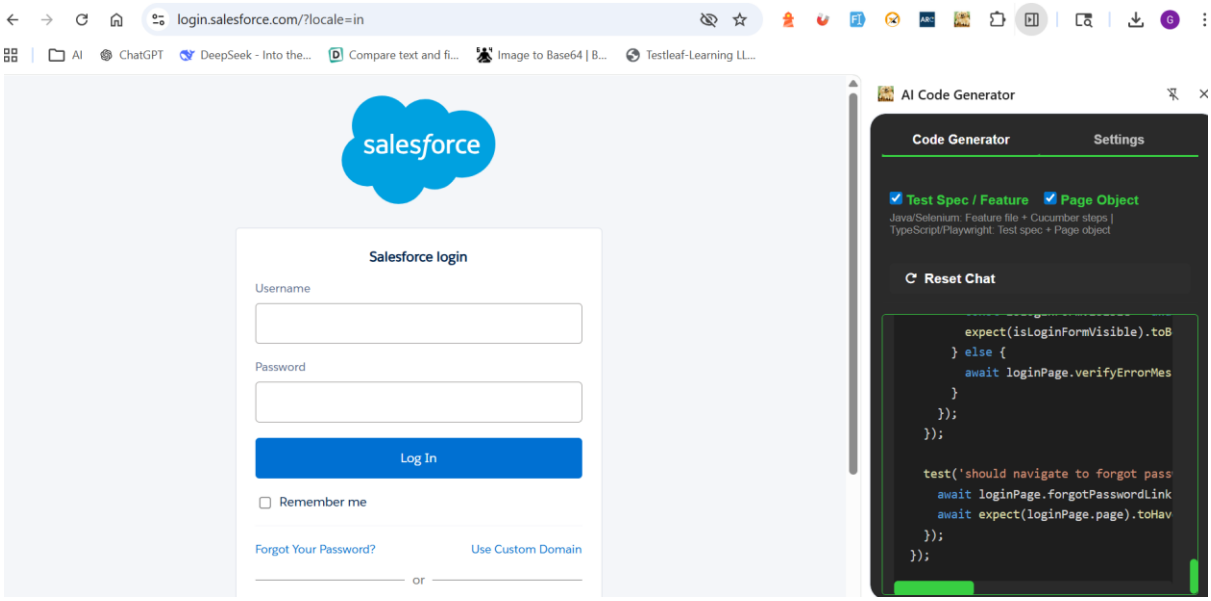

Code base:

```
// pages/login.page.ts
import { Page, Locator } from '@playwright/test';

export class LoginPage {
  readonly page: Page;
  readonly usernameField: Locator;
  readonly passwordField: Locator;
  readonly loginButton: Locator;
  readonly rememberMeCheckbox: Locator;
  readonly forgotPasswordLink: Locator;
  readonly errorMessage: Locator;

  constructor(page: Page) {
    this.page = page;
```

```
    this.usernameField = page.getByLabel('Username');
    this.passwordField = page.getByLabel('Password');
    this.loginButton = page.getByRole('button', { name: 'Log In' });
    this.rememberMeCheckbox = page.getByLabel('Remember me');
    this.forgotPasswordLink = page.getByRole('link', { name: 'Forgot Your
Password?' });
    this.errorMessage =
page.locator('#username_container').getByRole('alert');
  }

  async goto() {
    await this.page.goto('https://login.salesforce.com/?locale=in');
  }

  async login(username: string, password: string, rememberMe = false) {
    await this.usernameField.fill(username);
    await this.passwordField.fill(password);
    if (rememberMe) await this.rememberMeCheckbox.check();
    await this.loginButton.click();
  }

  async verifyErrorMessage(text: string) {
    await this.errorMessage.waitFor({ state: 'visible' });
    await expect(this.errorMessage).toContainText(text);
  }
}

// tests/login.spec.ts
import { test, expect } from '@playwright/test';
import { LoginPage } from '../pages/login.page';

const testUsers = [
  {
    username: 'suresh.sales@chennai.salesforce.com',
    password: 'Chennai@123',
    shouldSucceed: true,
    rememberMe: true
  },
  {
    username: 'lakshmi.marketing@bangalore.salesforce.com',
    password: 'Bangalore@456',
    shouldSucceed: true,
    rememberMe: false
  },
  {
    username: 'ramesh.admin@kerala.salesforce.com',
    password: 'Wrong@123',
    shouldSucceed: false,
    errorMessage: 'Please check your username and password. If you still
have trouble signing in, contact your Salesforce administrator.'
  },
  {
    username: 'invalid@tamilnadu.salesforce.com',
    password: 'Invalid@456',
    shouldSucceed: false,
    errorMessage: 'Please check your username and password. If you still
have trouble signing in, contact your Salesforce administrator.'
  },
  {
    username: 'kiran.operations@karnataka.salesforce.com',
    password: 'Karnataka@789',
```

```
    shouldSucceed: true,
    rememberMe: true
  }
];

test.describe('Salesforce Login Tests', () => {
  let loginPage: LoginPage;

  test.beforeEach(async ({ page }) => {
    loginPage = new LoginPage(page);
    await loginPage.goto();
  });

  test.describe.each(testUsers)('Login Scenarios', ({ username, password,
shouldSucceed, rememberMe, errorMessage }) => {
    test(`should ${shouldSucceed ? 'login successfully' : 'show error'} for
${username}`, async () => {
      await loginPage.login(username, password, rememberMe);

      if (shouldSucceed) {
        // Verify successful login by checking absence of error and form
submission
        const isLoginFormVisible = await loginPage.loginButton.isVisible();
        expect(isLoginFormVisible).toBe(false);
      } else {
        await loginPage.verifyErrorMessage(errorMessage);
      }
    });
  });

  test('should navigate to forgot password page', async () => {
    await loginPage.forgotPasswordLink.click();
    await expect(loginPage.page).toHaveURL(/\/secur\/forgotpassword\.jsp/);
  });
});
```